

Feedback System Performance

These slides are based on materials from the following books:

- J. Van de Vegte. *Feedback Control Systems*. Prentice-Hall, 1986.
- W. Bolton. *Mechatronics: A Multidisciplinary Approach*. 4th Edition, Pearson, 2008.

Objectives

When you have finished this lecture you should be able to:

- Understand the motivation of the use of feedback in mechatronic systems and its effect on performance.

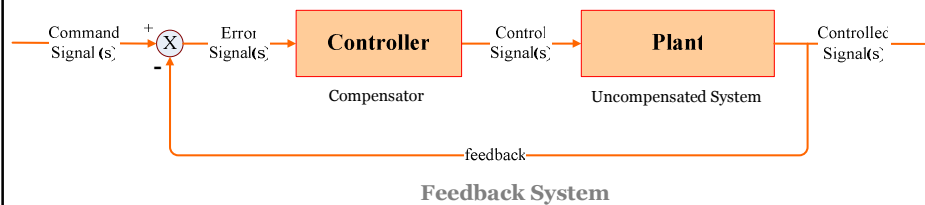
Outline

- Introduction
- Sensitivity to Parameter Variations
- Effect of External Disturbances
- Steady-State Errors
- Transient Response and Stability
- Transient Response vs Steady-State Errors
- Summary

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- **Introduction**
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Introduction



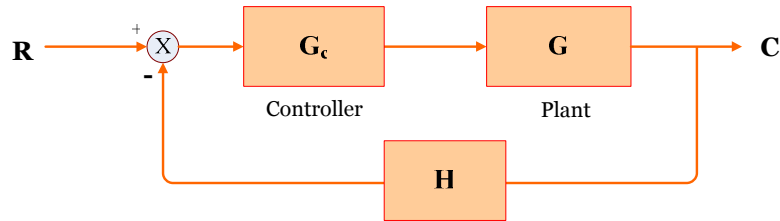
Motivations for feedback:

- Reducing the effects of parameter variations;
 - Reducing the effects of disturbance inputs;
 - Improving transient response characteristics
 - Reducing steady-state errors.
- } Prime reasons why feedback is needed

Outline

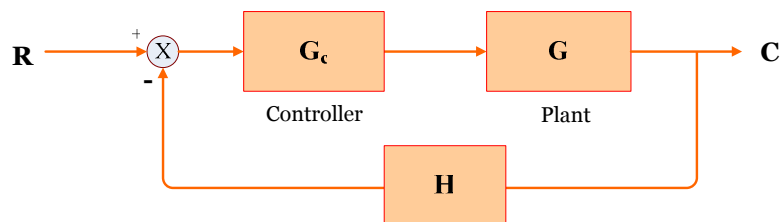
- Introduction
- **Sensitivity to Parameter Variations**
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Sensitivity to Parameter Variations



- ◇ The plant model **G** is usually an approximation to the actual dynamic behavior, and even then the parameter values in the model are often not precisely known and may also vary widely with the operating conditions.
- ◇ For very wide parameter variations, **adaptive** control schemes, which adjust the controller parameters, may be necessary.
- ◇ A prime advantage of feedback is that it can provide a strong reduction of the sensitivity without such changes of **G_c**.

Sensitivity to Parameter Variations



The closed-loop transfer function is:

$$T = \frac{C}{R} = \frac{G_c G}{1 + G_c G H}$$

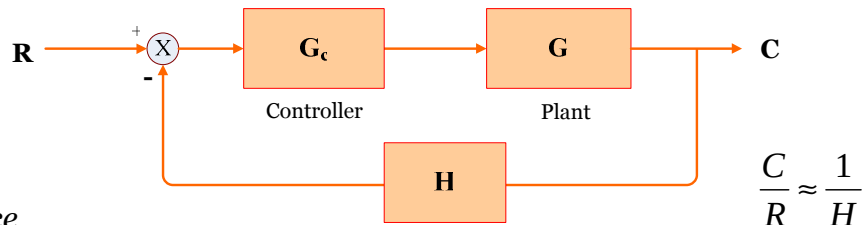
Let

$G_c G H$ = loop gain function (i.e., the product of the transfer functions around the loop)

If $G_c G H \gg 1$,

$$T = \frac{C}{R} \approx \frac{G_c G}{G_c G H} = \frac{1}{H}$$

Sensitivity to Parameter Variations



Hence

If the loop gain $G_c G H \gg 1$, C/R depends almost entirely on the feedback H alone, and is **virtually independent** of the plant and other elements in the forward path and of the variations of their parameters.

This is a major reason for the use of the feedback. The feedback H must be chosen for **small parameter variations**, but unlike G , it is usually under the control of the designer.

Sensitivity to Parameter Variations

Example: To illustrate how feedback can reduce the effects of parameter variations, consider the following system, where the blocks are simple gains.

Without feedback



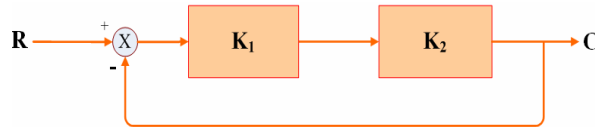
$$\frac{C}{R} = K_1 K_2$$

If the value of K_2 is halved, then C/R is halved also.

Sensitivity to Parameter Variations

Example (Cont'd):

With the feedback loop present



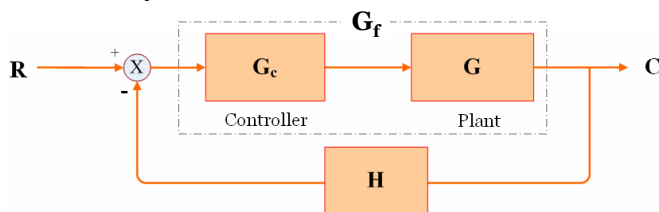
$$\frac{C}{R} = \frac{K_1 K_2}{1 + K_1 K_2} = T$$

- ◇ For $K_1 K_2 = 1 \rightarrow T = 1/2$, and
If $K_2 \rightarrow K_2/2$ then $T_{\text{new}} = 1/3$ or $T_{\text{new}} = 67\% T_{\text{original}}$
- ◇ But if $K_1 K_2 = 9$, then $T = 0.9$
If $K_2 \rightarrow K_2/2$ then $T_{\text{new}} = 0.818$ or $T_{\text{new}} = 91\% T_{\text{original}}$
- ◇ The larger loop gain has reduced the sensitivity.

Sensitivity to Parameter Variations

• Sensitivity Function

Formally, sensitivity properties can be studied by the use of the sensitivity function S .



where G_f = forward path transfer function

$$S = \frac{\partial T / T}{\partial G_f / G_f} = \frac{G_f}{T} \frac{\partial}{\partial G_f} \frac{G_f}{1 + G_f H}$$

$$= \frac{G_f}{T} \left(\frac{1}{1 + G_f H} - \frac{G_f H}{(1 + G_f H)^2} \right) = \frac{G_f}{T} \frac{1}{(1 + G_f H)^2} = \frac{1}{1 + G_f H}$$

Sensitivity to Parameter Variations

- **Sensitivity Function**

$$S = \frac{1}{1 + G_f H}$$

Static sensitivity: is the value of S for $s \rightarrow 0$

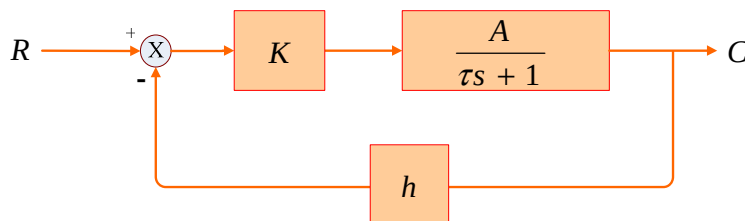
Dynamic sensitivities: are usually calculated by replacing s by $j\omega$ and plotting S as a function of frequency ω . Such results shows how sensitivity changes with the frequency of a sinusoidal input R .

Sensitivity to Parameter Variations

- **Example**

In the following figure the closed-loop transfer function is

$$T = \frac{KA}{\tau s + 1 + KA h}$$



The sensitivities S_A , S_h , and S_τ for small changes of A , h and τ are:

Sensitivity to Parameter Variations

• Example

$$S_A = \frac{\partial T / T}{\partial A / A} = \frac{A}{T} \frac{\partial}{\partial A} \frac{KA}{(\tau s + 1 + KAh)} = \frac{A}{T} \frac{K(\tau s + 1)}{(\tau s + 1 + KAh)^2} = \frac{\tau s + 1}{\tau s + 1 + KAh}$$

$$S_h = \frac{\partial T / T}{\partial h / h} = \frac{h}{T} \frac{\partial}{\partial h} \frac{KA}{(\tau s + 1 + KAh)} = \frac{h}{T} \frac{-(KA)^2}{(\tau s + 1 + KAh)^2} = \frac{-KAh}{\tau s + 1 + KAh}$$

$$S_\tau = \frac{\partial T / T}{\partial \tau / \tau} = \frac{\tau}{T} \frac{\partial}{\partial \tau} \frac{KA}{(\tau s + 1 + KAh)} = \frac{\tau}{T} \frac{-KAs}{(\tau s + 1 + KAh)^2} = \frac{-\tau s}{\tau s + 1 + KAh}$$

Notes:

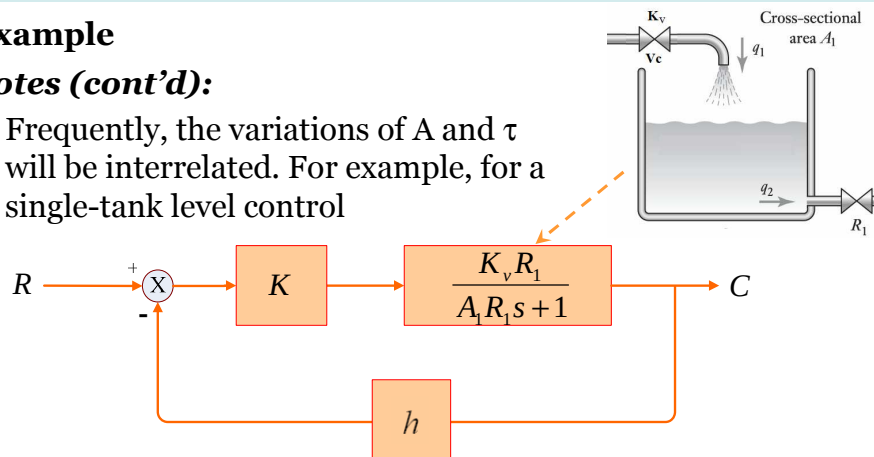
- ◇ The static sensitivity S_{As} is seen to reduce as loop gain KAh increases, but the magnitude of S_{hs} will approach 1.
- ◇ It is seen also that the static sensitivity $S_{\tau s}$ is zero

Sensitivity to Parameter Variations

• Example

Notes (cont'd):

- ◇ Frequently, the variations of A and τ will be interrelated. For example, for a single-tank level control



$$A = K_v R_1, \quad \tau = A_1 R_1$$

where A_1 and R_1 are the tank area and outlet valve resistance, K_v is the control valve gain

Sensitivity to Parameter Variations

• Example

Notes (cont'd):

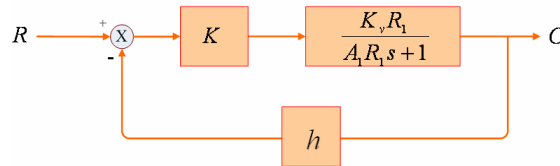
- ◇ Assuming that $K=9$, $K_v=1$, $A_1=1$.
- ◇ Leaving R_1 and h as free parameters, the closed-loop transfer function yields the unit step response

$$G = \frac{KK_v R_1}{A_1 R_1 s + 1} = \frac{9R_1}{R_1 s + 1}$$

$$\frac{C(s)}{R(s)} = \frac{G}{1 + GH} = \frac{9R_1 / (R_1 s + 1)}{1 + 9R_1 h / (R_1 s + 1)}$$

Taking inverse $L.T$ gives

$$c(t) = \frac{9}{9h + 1/R_1} \{1 - \exp[-(9h + 1/R_1)t]\}$$



Sensitivity to Parameter Variations

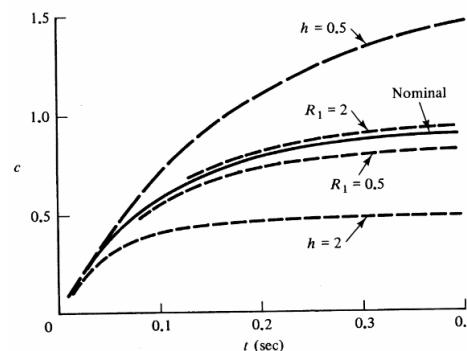
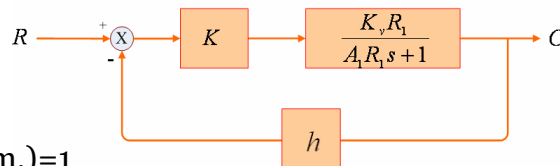
• Example

Notes (cont'd):

- ◇ Assuming that $R_1(\text{nom.})=1$ and $h(\text{nom.})=1$

$$c(t) = \frac{9}{9h + 1/R_1} \{1 - \exp[-(9h + 1/R_1)t]\}$$

- ◇ As Expected, the response is far more sensitive to h than to R_1 .

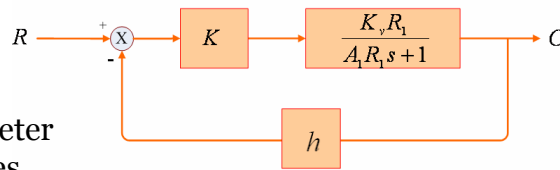


Sensitivity to Parameter Variations

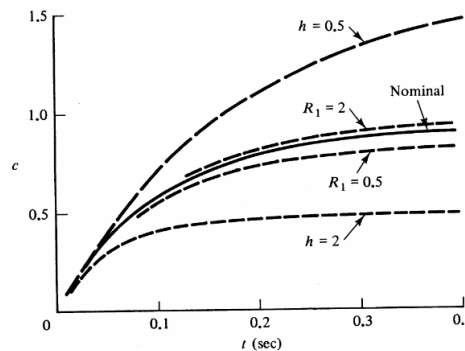
- **Example**

Notes (cont'd):

- ◇ In view of large parameter changes, the differences between the curves $R_1=1,2$ and 0.5 appear small.
- ◇ If K were reduced, reducing loop gain, these differences would be larger. This trend is evident in the curves in that loop gain is small for $h=0.5$ and $R_1=0.5$. Accordingly, these responses differ more for $h=2$ and $R_1=2$



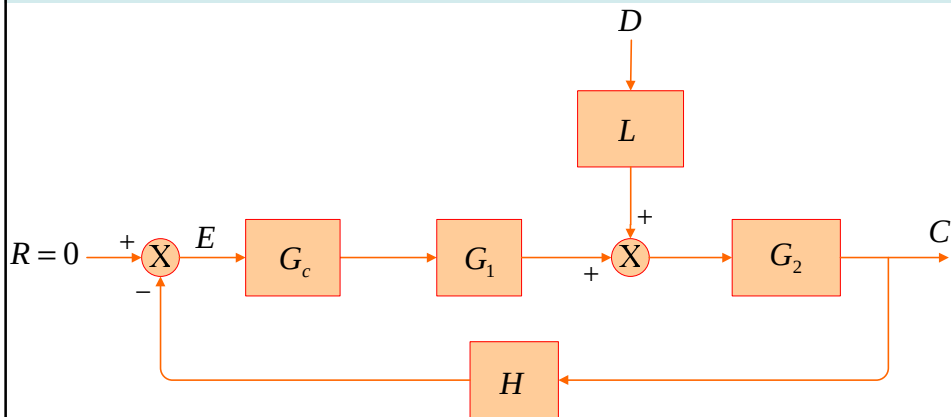
$$c(t) = \frac{9}{9h+1/R_1} \{1 - \exp[-(9h+1/R_1)t]\}$$



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Effect of External Disturbances



The effect of disturbance D on output C is given by the transfer function

$$\frac{C}{D} = \frac{G_2 L}{1 + G_c G_1 G_2 H} = -\frac{E}{D} \quad \text{for } H = 1$$

Effect of External Disturbances

$$\frac{C}{D} = \frac{G_2 L}{1 + G_c G_1 G_2 H} = -\frac{E}{D} \quad \text{for } H = 1$$

This shows the following:

◇ If loop gain $G_c G_1 G_2 \gg 1$, then

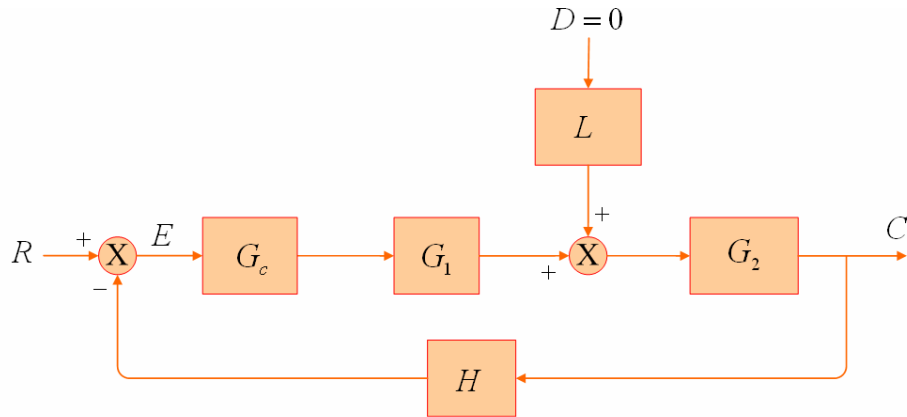
$$\frac{C}{D} \approx \frac{L}{G_c G_1 H} \quad \left(\approx -\frac{E}{D} \quad \text{for } H = 1 \right)$$

◇ So feedback strongly reduces the effect of a disturbance D on C if the loop gain is much greater than 1 due to high gain in the feedback path between C and D .

◇ This also shows why a system may respond well to an input R while the response to another input D is small.

Effect of External Disturbances

- For $H=1$, when $C/R=1$ is desired, loop gains $G_c G_1 G_2 \gg 1$ will ensure that $C/R = G_c G_1 G_2 / (1 + G_c G_1 G_2)$ is near 1.



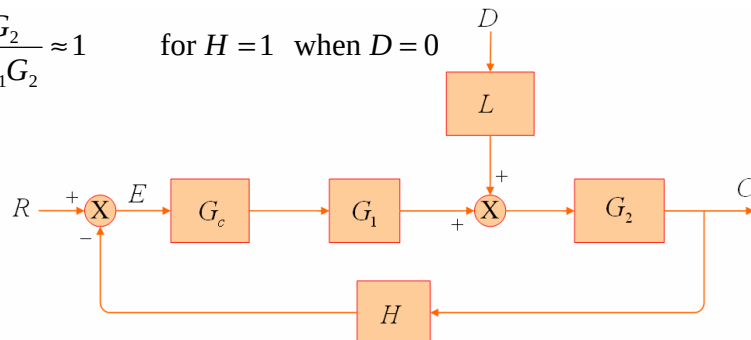
Effect of External Disturbances

- By locating the high gain in $G_c G_1$, between the points where R and D enter the loop, both requirements can met.

If loop gain $G_c G_1 G_2 H \gg 1$

$$\frac{C}{D} \approx \frac{L}{G_c G_1 H} \quad \left(\approx \frac{-E}{D} \quad \text{for } H=1 \right) \quad \text{when } R=0$$

$$\frac{C}{R} = \frac{G_c G_1 G_2}{1 + G_c G_1 G_2} \approx 1 \quad \text{for } H=1 \quad \text{when } D=0$$



Effect of External Disturbances

- ◇ Disturbances may also enter the feedback path, due to the sensor measuring C.
- ◇ The importance of measures to avoid this is clear, since normally there will not be a high gain in the feedback between C and such disturbance to attenuate their effect.

Effect of External Disturbances

- **Steady-state response**

The steady-state response, for $t \rightarrow \infty$, to a disturbance is clearly an important measure of system quality, and is readily found using the **final value theorem**.

For a unit step input $D=1/s$,

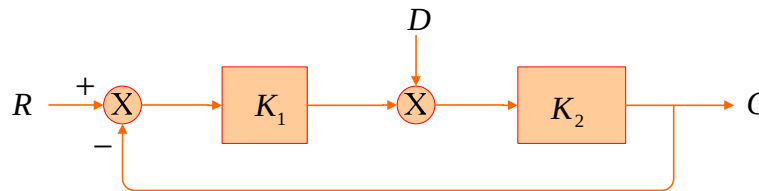
$$\lim_{t \rightarrow \infty} c(t) = \lim_{s \rightarrow 0} sC(s) = \lim_{s \rightarrow 0} \frac{G_2 L}{1 + G_c G_1 G_2 H}$$

Effect of External Disturbances

- **Steady-state response**

Example-1: In the following systems, R is the input which the output C should follow as closely as possible, and D is a disturbance input to which the output should ideally not respond at all. The blocks are simple gains. It is desired to find K_1 and K_2 for two sets of specifications:

- i) $C/D=0.1, C/R=0.9$
- ii) $C/D=0.01, C/R=0.99$



Effect of External Disturbances

- **Steady-state response**

Example-1 (cont'd): The transfer functions are

$$\frac{C}{R} = \frac{K_1 K_2}{\Delta}, \quad \frac{C}{D} = \frac{K_2}{\Delta}, \quad \Delta = 1 + K_1 K_2$$

and the solutions are readily found by substituting the specifications into these equations:

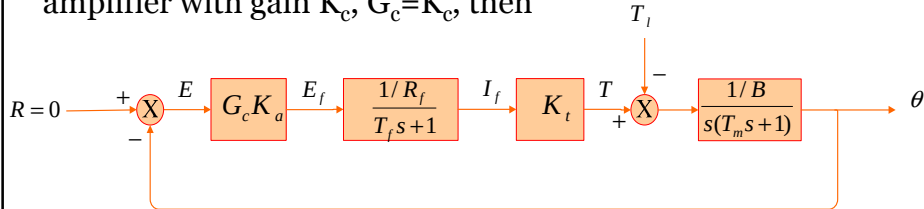
- i) $K_1 = 9, \quad K_2 = 1$
- ii) $K_1 = 99, \quad K_2 = 1$

Note that the more severe specifications (ii) require a large increase of gain between the points where R and D enter the loop, that is, in the forward path for R and in the feedback path for D .

Effect of External Disturbances

• Steady-state response

Example-2: For the motor position control system, if G_c is an amplifier with gain K_c , $G_c=K_c$, then



$$G_c G_1 = \frac{K_c K_a K_t / R_f}{T_f s + 1}, \quad L = H = 1, \quad G_2 = \frac{1/B}{s(T_m s + 1)}$$

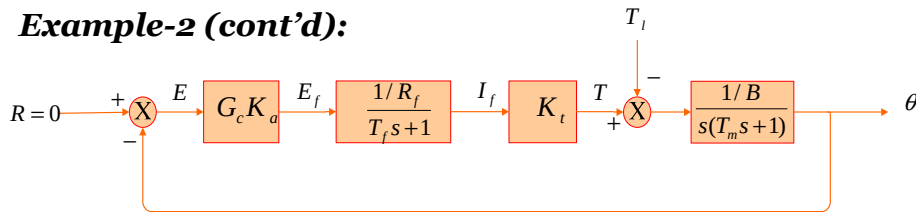
Hence, with negative sign at the load disturbance torque $T_l=1/s$,

$$\lim_{t \rightarrow \infty} c(t) = \lim_{s \rightarrow 0} \frac{-(T_f s + 1)/B}{s(T_m s + 1)(T_f s + 1) + K_c K_a K_t / (R_f B)} = -\frac{R_f}{K_c K_a K_t}$$

Effect of External Disturbances

• Steady-state response

Example-2 (cont'd):



$$\lim_{t \rightarrow \infty} c(t) = -\frac{R_f}{K_c K_a K_t}$$

Verification

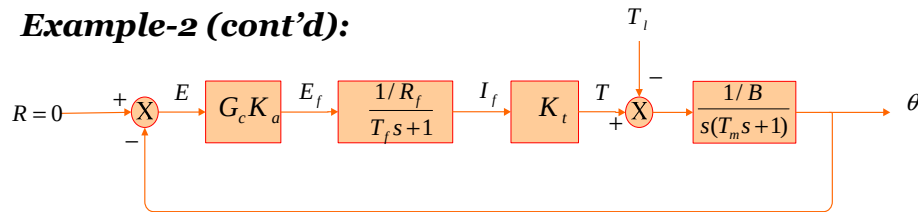
In steady state, the net torque $T-T_l$ must equal zero, so if $T_l=1$, T must be 1 as well. From the diagram, this requires that

$$E = T(T_f s + 1) \frac{R_f}{K_c K_a K_t} \quad \text{at s.s.} \quad e = 1(0+1) \frac{R_f}{K_c K_a K_t} = \frac{R_f}{K_c K_a K_t}$$

Effect of External Disturbances

- Steady-state response

Example-2 (cont'd):



$$\lim_{t \rightarrow \infty} c(t) = -\frac{R_f}{K_c K_a K_t}$$

Verification (cont'd)

Since input r is zero while considering the response to T_l

$$e = r - c = -c$$

$$c = -r = -\frac{R_f}{K_c K_a K_t}$$

Effect of External Disturbances

- Steady-state response

Example-2 (cont'd):

$$\lim_{t \rightarrow \infty} c(t) = -\frac{R_f}{K_c K_a K_t} \quad \Rightarrow \quad \text{Steady-state response to a the load disturbance torque}$$

High controller gain, therefore, not only makes the system less sensitive to parameter variations but also improves its stiffness against load disturbance.

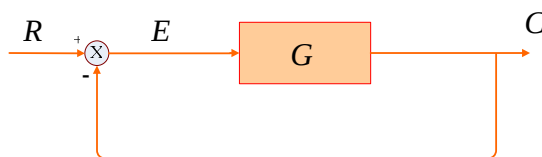
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Steady-State Errors

High loop gains are also desirable from the point of view of the reduction of steady-state errors in feedback systems.

Consider the following unity feedback system:



$$\begin{aligned} e_{ss} &= \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) \\ &= \lim_{s \rightarrow 0} s[R(s) - C(s)] \\ &= \lim_{s \rightarrow 0} s \left[R(s) - \frac{R(s)}{1 + G(s)} \right] = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)} \end{aligned}$$

Steady-State Errors

For $G(s)$, the following general form is assumed:

$$G(s) = \frac{K}{s^n} \frac{a_k s^k + \dots + a_1 s + 1}{b_l s^l + \dots + b_1 s + 1}$$

In this equation:

- ◇ **Gain:** K as given, with the constant terms in numerator and denominator polynomials made unity, is formally the gain of the transfer function G .
- ◇ **Type number:** the type number of G is the value of integer n (number of integrators in G)

Examples 

$$G(s) = \frac{K_1(as+b)}{(cs^2+ds+e)(fs+g)} : \Rightarrow \text{type 0}$$

$$G(s) = \frac{K_1(as+b)}{s(cs^2+ds+e)(fs+g)} : \Rightarrow \text{type 1}$$

Steady-State Errors

For $G(s)$, the following general form is assumed:

$$G(s) = \frac{K}{s^n} \frac{a_k s^k + \dots + a_1 s + 1}{b_l s^l + \dots + b_1 s + 1}$$

In this equation:

- ◇ **Gain** $K = \lim_{s \rightarrow 0} s^n G(s)$ and a common practice associates the following names and notations with K , depending on n :

$n=0$:	K_p =position error constant
$n=1$:	K_v =velocity error constant
$n=2$:	K_a =acceleration error constant

Steady-State Errors

$$G(s) = \frac{K}{s^n} \frac{a_k s^k + \dots + a_1 s + 1}{b_l s^l + \dots + b_1 s + 1}$$

Examples:

$$G(s) = \frac{K_1(as+b)}{(cs^2+ds+e)(fs+g)} : \text{type 0} \quad \text{gain } K_p = \frac{K_1b}{eg}$$

$$G(s) = \frac{K_1(as+b)}{s(cs^2+ds+e)(fs+g)} : \text{type 1} \quad \text{gain } K_v = \frac{K_1b}{eg}$$

Steady-State Errors

$$K = \lim_{s \rightarrow 0} s^n G(s)$$

The above equation shows that

$$\lim_{s \rightarrow 0} G(s) = \lim_{s \rightarrow 0} (K / s^n)$$

so that the steady-state error equation can be written as follows:

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + (K / s^n)}$$

	Type Number: $n = 0$	$n = 1$	$n = 2$
Step $u(t)$; $R = 1/s$	$\frac{1}{1 + K_p}$	0	0
Ramp t ; $R = 1/s^2$	∞	$\frac{1}{K_v}$	0
Acceleration $t^2/2$; $R = 1/s^3$	∞	∞	$\frac{1}{K_a}$

Steady-State Errors

- ◇ The nonzero finite errors decrease **as gain is increased**, as do parameter sensitivity and disturbance response.

Type Number:	$n = 0$	$n = 1$	$n = 2$
Step $u(t)$; $R = 1/s$	$\frac{1}{1 + K_p}$	0	0
Ramp t ; $R = 1/s^2$	∞	$\frac{1}{K_v}$	0
Acceleration $t^2/2$; $R = 1/s^3$	∞	∞	$\frac{1}{K_a}$

Steady-State Errors

- ◇ For insight into these results it is useful to consider the following systems:

Type 0 System

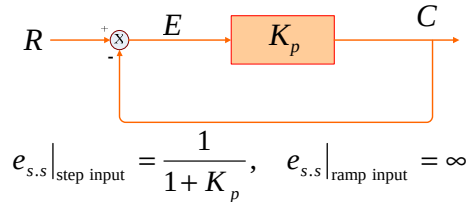
For $n=0$, $C=K_p E$,

$$c(t)=K_p e(t)$$

so there cannot be a nonzero output without a proportional, nonzero error.

If the output must increase along a ramp, the error must increase according to a ramp as well.

So a type 0 system has a steady-state error for a step input and cannot follow a ramp.

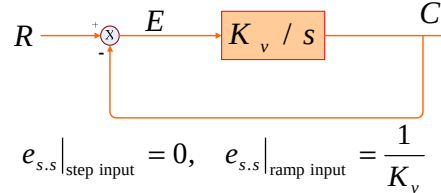


Steady-State Errors

Type 1 System

For $n=1$, $C=K_v/s \cdot E$ so

$$c(t) = K_v \int e(t) dt$$



This means that the output cannot level off to a constant value unless the error levels off at zero.

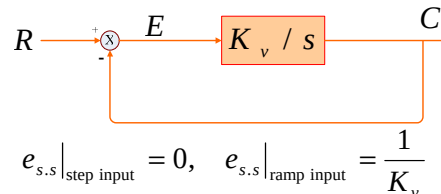
A steady-state with nonzero steady-state error cannot exist for a step input because the integrator would cause the output to change.

Steady-State Errors

Type 1 System

For $n=1$, $C=K_v/s \cdot E$ so

$$c(t) = K_v \int e(t) dt$$



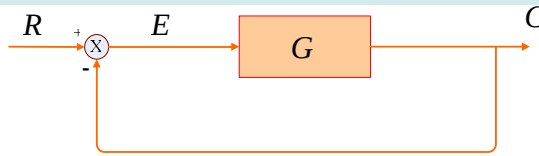
From a design point of view, this means that if, as it is often the case, the performance specifications require zero steady-state error after a step input, the designer must ensure that the system is at least of type 1.

Type 2 system would be necessary if zero steady-state errors following both steps and ramps are specified.

$$\text{For type 2, } e_{s.s}|_{step\ input} = 0, \quad e_{s.s}|_{ramp\ input} = 0$$

Steady-State Errors

Example-1:



(a) $G(s) = \frac{K_1(as+b)}{(cs^2+ds+e)(fs+g)}$: type 0
gain $K_p = \frac{K_1b}{eg}$

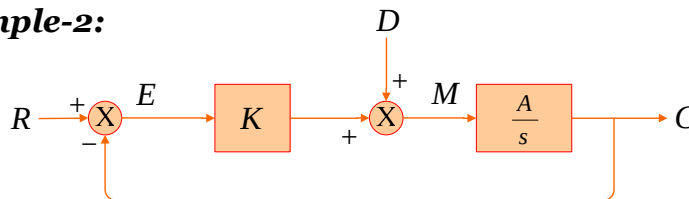
Unit step : $e_{ss} = \frac{1}{1+K_p}$ unit ramp : $e_{ss} \rightarrow \infty$

(b) $G(s) = \frac{K_1(as+b)}{s(cs^2+ds+e)(fs+g)}$: type 1
gain $K_v = \frac{K_1b}{eg}$

Unit step : $e_{ss} = 0$ unit ramp : $e_{ss} = \frac{1}{K_v}$

Steady-State Errors

Example-2:

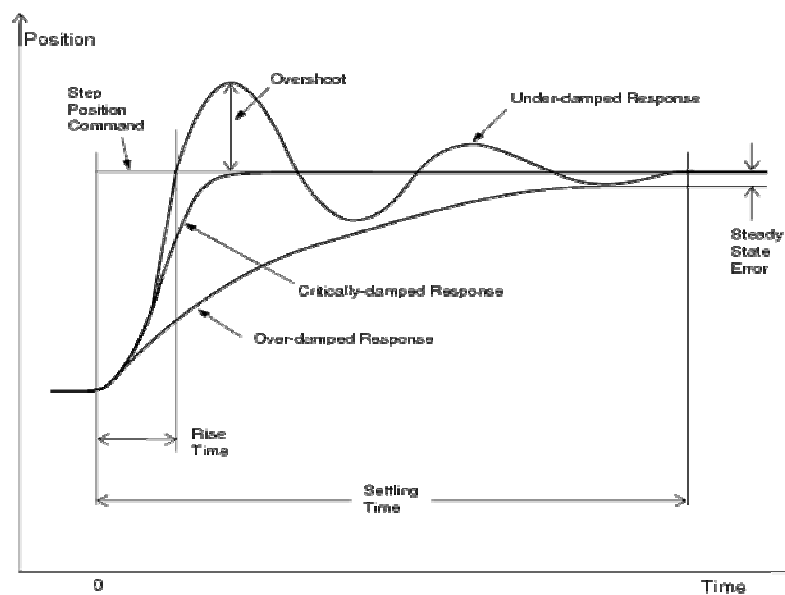


- ◇ The system type is 1, so $e_{ss}=0$ for a step input R.
- ◇ For input D, the steady-state value m_{ss} of M is also zero (since otherwise c_{ss} could not be constant), but the feedback from C to D is not unity but K.
- ◇ For a unit value of D the condition for $m_{ss}=0$ is $1+Ke_{ss}=0$, so $e_{ss}=-1/K$.
- ◇ Direct application of the final value theorem to $C(s)$ will verify these results.

Outline

- Introduction
- Sensitivity to Parameter Variations
- Effect of External Disturbances
- Steady-State Errors
- **Transient Response and Stability**
- Transient Response vs Steady-State Errors
- Summary

Transient Response and Stability



Transient Response and Stability

System Stability is the most important characteristic of the transient response.

A system is **stable** if the transient solution decays to zero and is unstable if this solution grows.

A stable system is one that produces a **bounded, or finite, response** when subjected to a **bounded input**.

Stability Theorem: A system is stable if and only if all system poles lie in the left half of the s-plane.

Transient Response and Stability

Stability Theorem: A system is stable if and only if all system poles lie in the left half of the s-plane.

The transfer function provides a basis for determining important system response characteristics without solving the complete differential equation. The transfer function is a rational function in the complex variable $s = \sigma + j\omega$, that is:

$$H(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}$$

$$H(s) = \frac{\text{numerator}(s)}{\text{denominator}(s)} = \frac{N(s)}{D(s)} = K \frac{(s - z_1)(s - z_2) \dots (s - z_{m-1})(s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_{n-1})(s - p_n)}$$

$$K = b_m / a_n.$$

Transient Response and Stability

• Poles and Zeros

$$H(s) = \frac{N(s)}{D(s)} = K \frac{(s - z_1)(s - z_2) \dots (s - z_{m-1})(s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_{n-1})(s - p_n)}$$

$z_i \gg$ system zeros or roots of the equation $N(s) = 0$

$p_i \gg$ system poles or roots of the equation $D(s) = 0$

$D(s)=0$ is called the characteristic equation.

All of the coefficients of polynomials $N(s)$ and $D(s)$ are real, therefore the poles and zeros must be either purely real, or appear in complex conjugate pairs.

Transient Response and Stability

• Poles and Zeros

Example-1:

A linear system is described by the differential equation

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 2\frac{du}{dt} + 1.$$

Find the system poles and zeros.

Solution: From the differential equation the transfer function is

$$H(s) = \frac{2s + 1}{s^2 + 5s + 6}.$$

which may be written in factored form

$$\begin{aligned} H(s) &= \frac{1}{2} \frac{s + 1/2}{(s + 3)(s + 2)} \\ &= \frac{1}{2} \frac{s - (-1/2)}{(s - (-3))(s - (-2))}. \end{aligned}$$

The system therefore has a single real zero at $s = -1/2$, and a pair of real poles at $s = -3$ and $s = -2$.

Transient Response and Stability

- **Poles and Zeros**

Example-2:

A system has a pair of complex conjugate poles $p_1, p_2 = -1 \pm j2$, a single real zero $z_1 = -4$, and a gain factor $K = 3$. Find the differential equation representing the system.

Solution: The transfer function is

$$\begin{aligned} H(s) &= K \frac{s - z}{(s - p_1)(s - p_2)} \\ &= 3 \frac{s - (-4)}{(s - (-1 + j2))(s - (-1 - j2))} \\ &= 3 \frac{(s + 4)}{s^2 + 2s + 5} \end{aligned}$$

and the differential equation is

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = 3\frac{du}{dt} + 12u$$

Transient Response and Stability

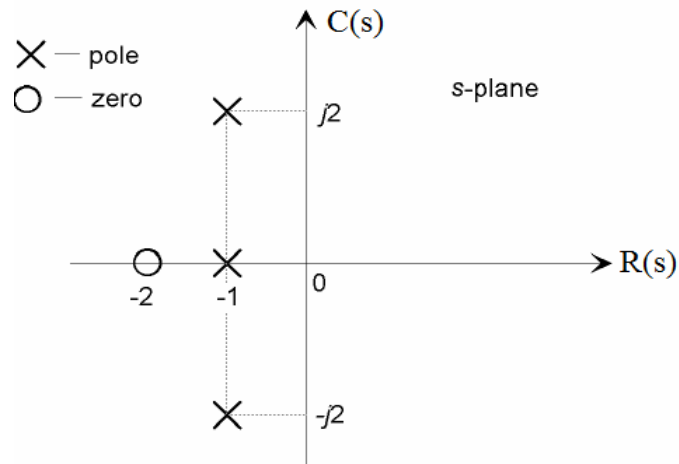
- **Poles and Zeros**

A system is characterized by its poles and zeros in the sense that they allow reconstruction of the input/output differential equation. In general, the poles and zeros of a transfer function may be complex, and the system dynamics may be represented graphically by plotting their locations on the complex s -plane, whose axes represent the real and imaginary parts of the complex variable s . Such plots are known as **pole-zero plots**. The location of the poles and zeros provide qualitative insights into the response characteristics of a system.

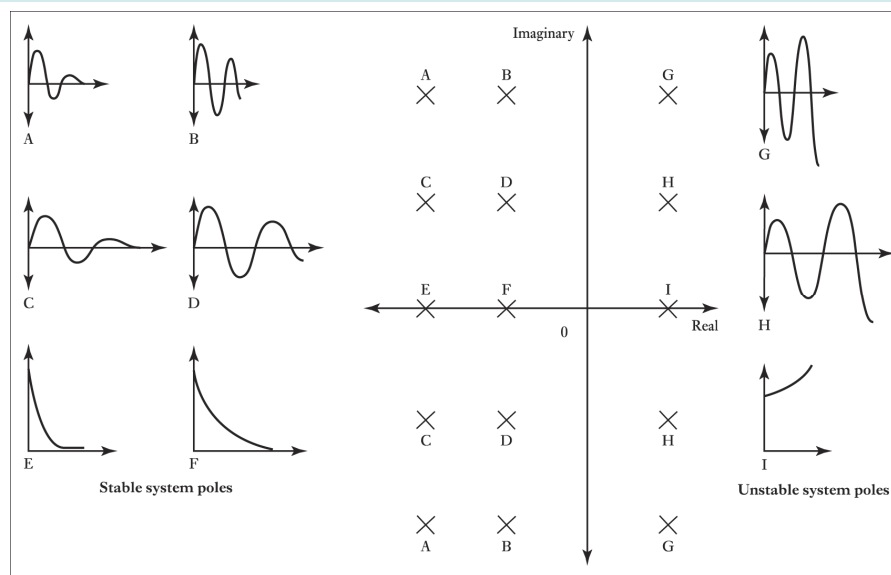
Transient Response and Stability

• Poles and Zeros

$$H(s) = \frac{(3s+6)}{(s^3+3s^2+7s+5)} = 3 \frac{(s-(-2))}{(s-(-1))(s-(-1-2j))(s-(-1+2j))}$$



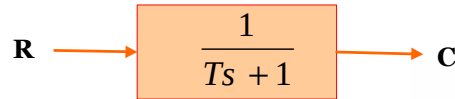
Transient Response and Stability



The s-Plane

Transient Response and Stability

• Simple Lag: First-Order Systems



$$C(s) = \frac{1/T}{s(s + 1/T)} = \frac{K_1}{s} + \frac{K_2}{s + 1/T}$$

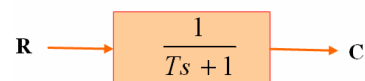
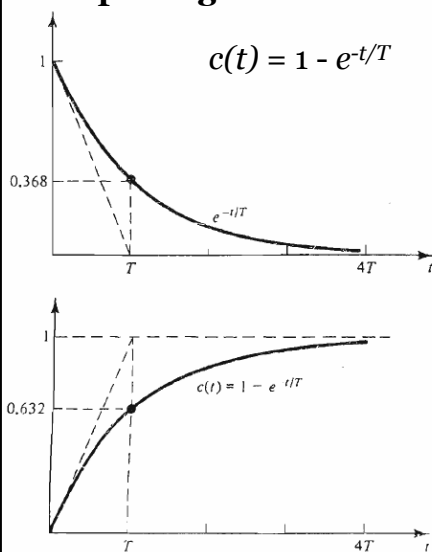
$$K_1 = \left. \frac{1/T}{s + 1/T} \right|_{s=0} = 1 \quad K_2 = \left. \frac{1/T}{s} \right|_{s=-1/T} = -1$$

Hence the transient response is $c(t) = 1 - e^{-t/T}$

The first term is the forced solution, due to the input, and the second the transient solution, due to the system pole.

Transient Response and Stability

• Simple Lag: First-Order Systems



Note that: The transient is seen to be a decaying exponential. If it takes long to decay, the system response is slow, so the speed of decay is of key importance. The commonly used measure of this speed of decay is the time constant.

Step response of simple lag

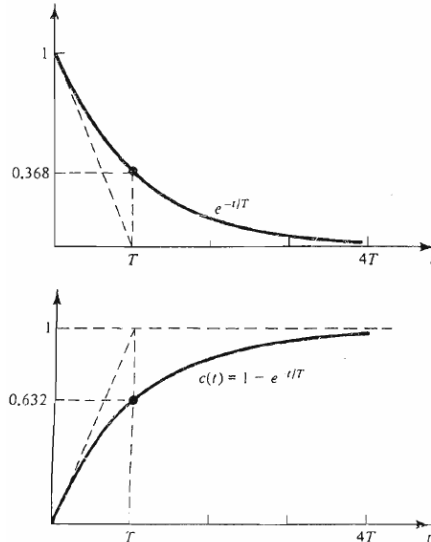
Transient Response and Stability

• Simple Lag: First-Order Systems

◇ **Time constant T:** This is the time in seconds for the decaying exponential transient to be reduced to $e^{-1} = 0.368$ of its initial value.

◇ **T** indicates the speed of decay.

◇ It takes $4T$ seconds for the transient to decay to 1.8% of its initial value.



Transient Response and Stability

• Simple Lag: First-Order Systems

For the simple lag, two features are important:

1. **Stability:** As discussed, for stability, the system pole $-1/T$ must lie in the left half of the s-plane, since otherwise the transient $e^{-t/T}$ grows instead of decays as t increases.
2. **Speed of response:** To speed up the response of the system (that is, to reduce its time constant T), the pole $-1/T$ must be moved left.

Transient Response and Stability

• Quadratic Lag: Second-Order Systems

This very common transfer function can always be reduced to the standard form:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

where ω_n = undamped natural frequency
 ζ = damping ratio

For a unit step input $R(s) = 1/s$, the transform of the output is

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

Transient Response and Stability

• Quadratic Lag: Second-Order Systems

There are three possibilities, depending on the roots of the system characteristic equation:

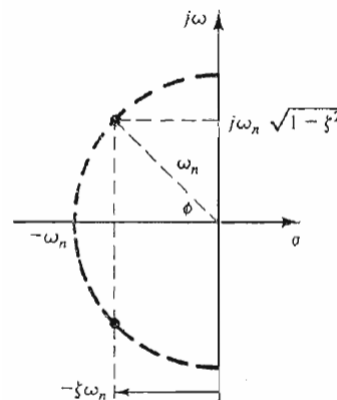
$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

These system poles depend on ζ :

$\zeta > 1$: overdamped : $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

$\zeta = 1$: critically damped : $s_{1,2} = -\omega_n$

$\zeta < 1$: underdamped : $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$



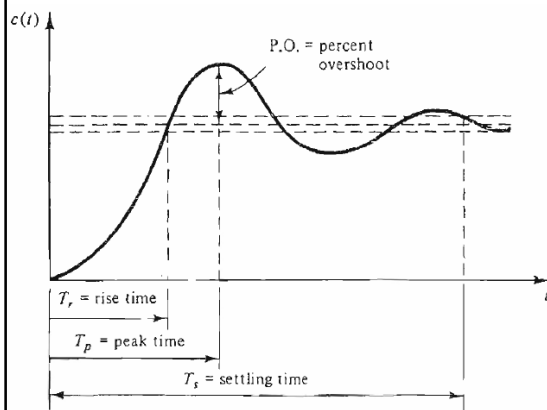
Transient Response and Stability

• Quadratic Lag: Second-Order Systems

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$\zeta = 1$: critically damped : $s_{1,2} = -\omega_n$

$\zeta < 1$: underdamped : $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$



$$T = \frac{1}{\zeta\omega_n}$$

$$T_s = 4T = \frac{4}{\zeta\omega_n}$$

$$T_p = 4T = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$P.O = 100 \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

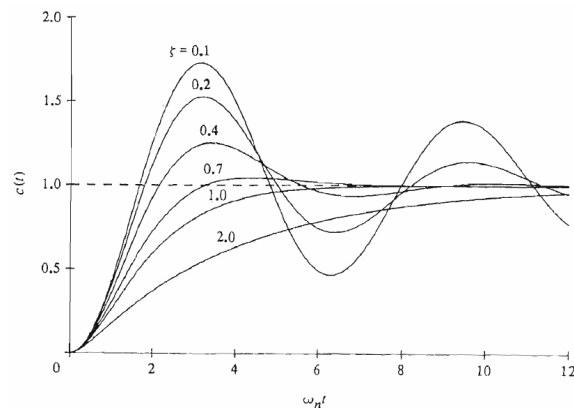
Transient Response and Stability

• Quadratic Lag: Second-Order Systems

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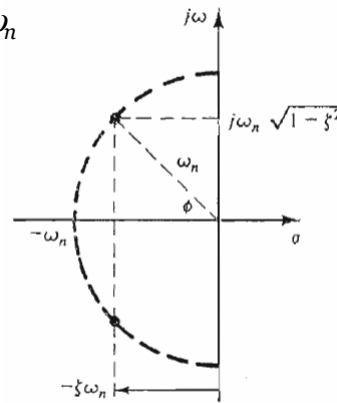


Transient Response and Stability

• Quadratic Lag: Second-Order Systems

As for the simple lag, it is again important to determine the correlations between dynamic behavior and the pole positions in the s-plane

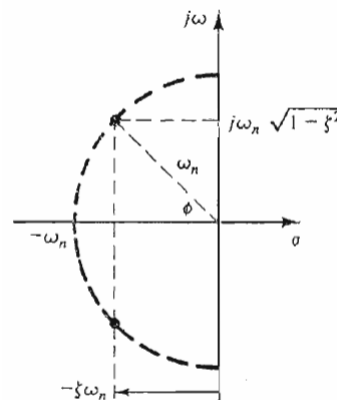
- ◇ **Absolute stability:** The real part $-\zeta\omega_n$ of the poles must be negative for the transient to decay; that is, the poles must lie in the left half of the s-plane.
- ◇ **Relative stability:** To avoid excessive overshoot and unduly oscillatory behavior, damping ratio ζ must be adequate. Since $\zeta = \cos \phi$, the angle ϕ may not be close to 90° .



Transient Response and Stability

• Quadratic Lag: Second-Order Systems

- ◇ **Time constant:** The time constant is reduced (that is, the speed of decay of the transient is increased) by increasing the negative real part of the pole positions.
- ◇ **Speed of response:** Noting that the horizontal axis in terms of wilt, it is seen that the speed of response of the system is increased by increasing the distance ω_n of the poles to the origin.

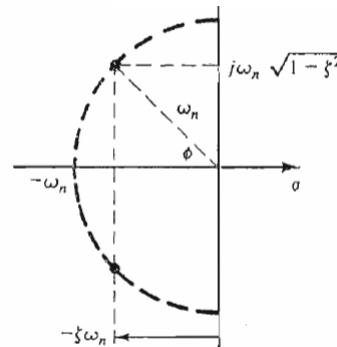


Transient Response and Stability

• Quadratic Lag: Second-Order Systems

- ◇ **Undamped natural frequency ω_n** : This equals the distance of the poles to the origin. Moving the poles out radially (that is, with ζ constant) increases the speed of response while the percentage overshoot remains constant.

- ◇ **Frequency of transient oscillations $\omega_n \sqrt{1 - \zeta^2}$** : This frequency, also called the resonant frequency or damped natural frequency, equals the imaginary part of the pole positions.



Transient Response and Stability

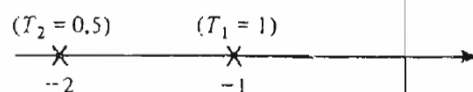
• Quadratic Lag: Second-Order Systems

Example for $\zeta > 1$:
$$G(s) = \frac{2}{s^2 + 3s + 2} = \frac{2}{(s+1)(s+2)}$$

For a unit step,
$$C(s) = \frac{2}{s(s+1)(s+2)} = \frac{K_1}{s} + \frac{K_2}{s+1} + \frac{K_3}{s+2}$$

and it is found that
$$e(t) = 1 - 2e^{-t} + e^{-2t}$$

The transient consists of exponentials with time constants $T_1=1$ and $T_2=0.5$, could have been predicted from inspection of the system pole-zero pattern. The system is a series connection of two simple lags. The **dominating pole** is the one at -1.



Transient Response and Stability

• Higher-Order Systems and Dominating Poles

The *total transient response* is a **superposition** of exponential decays and decaying oscillations.

- ◇ The dominating-poles concept simplifies both analysis and design greatly.
- ◇ The response of many systems is dominated by one pair of complex poles relatively close to the imaginary axis, so they behave approximately as second-order systems.
- ◇ The number of additional poles in near-dominant locations that can cause deviations from this second-order behavior is usually very limited.

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Absolute stability requires only that all roots of the system characteristic equation

$$a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0$$

(that is, the poles of its transfer function) lie in the left-half s-plane.

Routh-Hurwitz criterion provides a quick method for determining absolute, but not relative, stability from the coefficients, without calculating the roots. It shows how many system poles are in the right-half s-plane or on the imaginary axis.

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

The *Routh-Hurwitz criterion* states:

1. A necessary and sufficient condition for stability is that there be no changes of sign in the elements of the first column of the array.
2. The number of these sign changes is equal to the number of roots in the right-half s-plane.
3. If the first element in a row is zero, it is replaced by a very small positive number ε , and the sign changes when $\varepsilon \rightarrow 0$ are counted after completing the array.
4. If all elements in a row are zero, the system has poles in the right half-plane or on the imaginary axis.

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Routh Array:

$$a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0 \quad \Rightarrow$$

$s^n:$	a_n	a_{n-2}	a_{n-4}	$\dots$	0
	$\downarrow$	$\nearrow$	$\downarrow$	$\nearrow$	
$s^{n-1}:$	a_{n-1}	a_{n-3}	a_{n-5}	$\dots$	0
$s^{n-2}:$	b_1	b_2	b_3	$\dots$	0
$s^{n-3}:$	c_1	c_2	c_3	$\dots$	0
$\vdots$					
$s^1:$	g_1	0			
$s^0:$	h_1	0			

$$b_1 = \frac{1}{a_{n-1}}(a_{n-1}a_{n-2} - a_n a_{n-3})$$

$$b_2 = \frac{1}{a_{n-1}}(a_{n-1}a_{n-4} - a_n a_{n-5})$$

$$b_3 = \frac{1}{a_{n-1}}(a_{n-1}a_{n-6} - a_n a_{n-7})$$

$$c_1 = \frac{1}{b_1}(b_1 a_{n-3} - b_2 a_{n-1})$$

$$c_2 = \frac{1}{b_1}(b_1 a_{n-5} - b_3 a_{n-1})$$

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Example-1: $s^5 + s^4 + 6s^3 + 5s^2 + 12s + 20 = 0$

the Routh array is

$$\begin{array}{l} s^5: \quad 1 \quad \quad 6 \quad 12 \quad 0 \\ s^4: \quad 1 \quad \quad 5 \quad 20 \quad 0 \\ s^3: \quad 1 \quad \quad -8 \quad 0 \\ s^2: \quad 13 \quad \quad 20 \quad 0 \\ s^1: \quad \frac{-124}{13} \quad 0 \\ s^0: \quad 20 \quad \quad 0 \end{array}$$

There are two sign changes in the first column, first from plus to minus and then from minus to plus. Hence this characteristic equation represents an unstable system, with two poles in the right-half s-plane

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Example-2: $s^5 + s^4 + 5s^3 + 5s^2 + 12s + 10 = 0$

$$\begin{array}{l} s^5: \quad 1 \quad 5 \quad 12 \quad 0 \\ s^4: \quad 1 \quad 5 \quad 10 \quad 0 \\ s^3: \quad \varepsilon \quad 2 \quad 0 \\ s^2: \quad c_1 \quad 10 \quad 0 \\ s^1: \quad d_1 \quad 0 \\ s^0: \quad 10 \end{array}$$

The zero element in the first column for s^3 in the following Routh array is replaced by a small positive number ε . Then

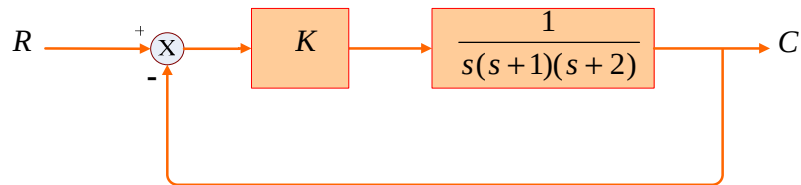
$$c_1 = \frac{5\varepsilon - 2}{\varepsilon} \approx \frac{-2}{\varepsilon} \quad d_1 = \frac{2c_1 - 10\varepsilon}{c_1} \approx 2$$

Thus c_1 is a large negative number, implying two unstable poles

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Example-3: For the electromechanical servo modeled in the following diagram, find the limits on K for stability.



$$\frac{C}{R} = \frac{K}{s^3 + 3s^2 + 2s + K}$$

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Example-3 (cont'd):

$$\frac{C}{R} = \frac{K}{s^3 + 3s^2 + 2s + K}$$

The characteristic equation is $s^3 + 3s^2 + 2s + K = 0$

Routh array is

s^3	:	1	2	0
s^2	:	3	K	0
s^1	:	$\frac{6-K}{3}$	0	
s^0	:	K	0	

Evidently, $0 < K < 6$ is the range of K for a stable system.

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Transient Response vs Steady-State Errors

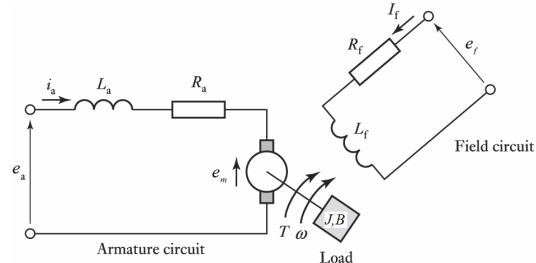
- ◇ It has been shown that parameter sensitivity, disturbance response, and steady-state errors are all improved by increased gain.
- ◇ Dynamic response considerations usually limit the permissible gain as transient response and accuracy are related.
- ◇ A higher gain means a larger signal to the actuator for a certain change of error. This suggests smaller error, but also increasing danger of overcorrection and oscillations.
- ◇ In fact, feedback design is generally concerned with achieving a satisfactory compromise between relative stability, that is, adequate damping ratio ζ of the dominating closed-loop poles, and accuracy, that is, small errors.

Transient Response vs Steady-State Errors

Example-1: Step Responses of a DC Motor Position Servo

let the transfer function from field voltage to shaft position of a field-controlled dc motor be

$$G(s) = \frac{0.5}{s(0.25s + 1)}$$



Assuming a fast response power amplifier and a constant-gain controller, the loop can be considered to be closed, as shown, by an amplifier block of gain K .

The closed-loop transfer function is:

$$\frac{C(s)}{R(s)} = \frac{2K}{s^2 + 4s + 2K}$$

Transient Response vs Steady-State Errors

Example-1: Step Responses of a DC Motor Position Servo

and for a unit step input $[R(s) = 1/s]$, the transform of the shaft position is:

$$C(s) = \frac{2K}{s(s^2 + 4s + 2K)}$$

- $K = 1$

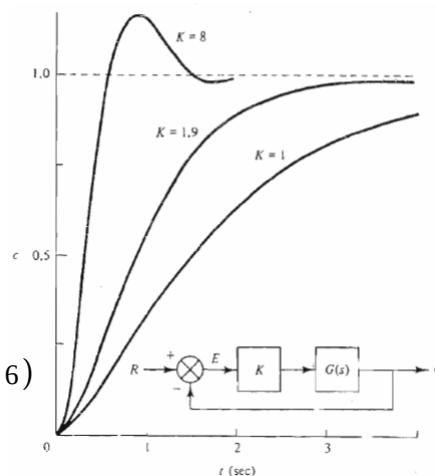
$$c(t) = 1 - 1.207 e^{-0.586 t} + 0.207 e^{-3.414 t}$$

- $K = 1.9$

$$c(t) = 1 - 2.737 e^{-1.553 t} + 1.737 e^{-2.447 t}$$

- $K = 8$

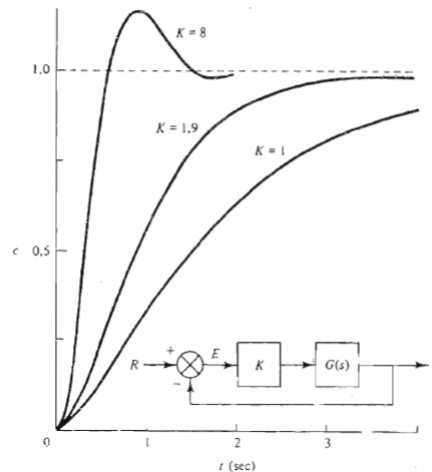
$$c(t) = 1 + 1.1547 e^{-2t} \cos(3.4641 t + 5\pi / 6)$$



Transient Response vs Steady-State Errors

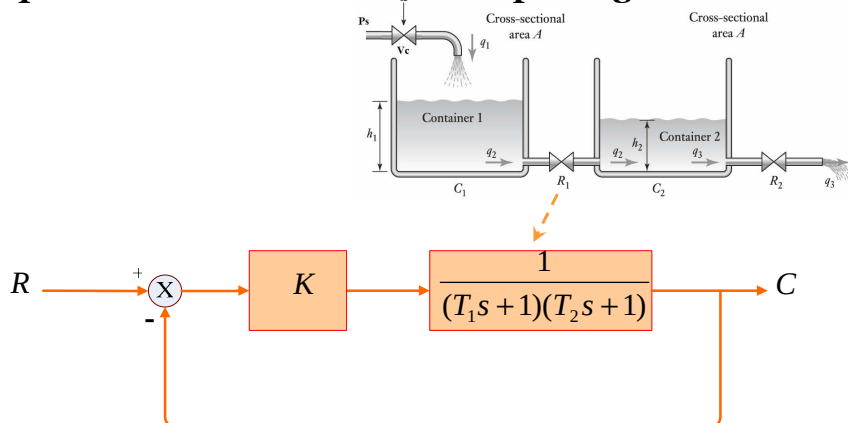
Example-1: Step Responses of a DC Motor Position Servo

- ◇ **Larger K produces faster response** for, because the motor input changes more for a given change of error, and a danger of **overcorrection and oscillations** if K is raised further.
- ◇ The steady-state or final value of the error is seen to be zero for all three values of K, as expected for this **type 1** system.



Transient Response vs Steady-State Errors

Example-2: P Control with Two Simple Lags



Among other possibilities, this system could represent a two-tank level control, a temperature control with two heat capacitances in series, or a motor speed control.

Transient Response vs Steady-State Errors

Example-2: P Control with Two Simple Lags

the system is type 0 with gain K , so the steady-state error after a unit step is $1/(1 + K)$

Consider now stability, using $T_1 = 1$, $T_2 = 0.5$ for a numerical example. The closed-loop transfer function and the system characteristic equation are, respectively,

$$\frac{C(s)}{R(s)} = \frac{2K}{s^2 + 3s + 2 + 2K}$$

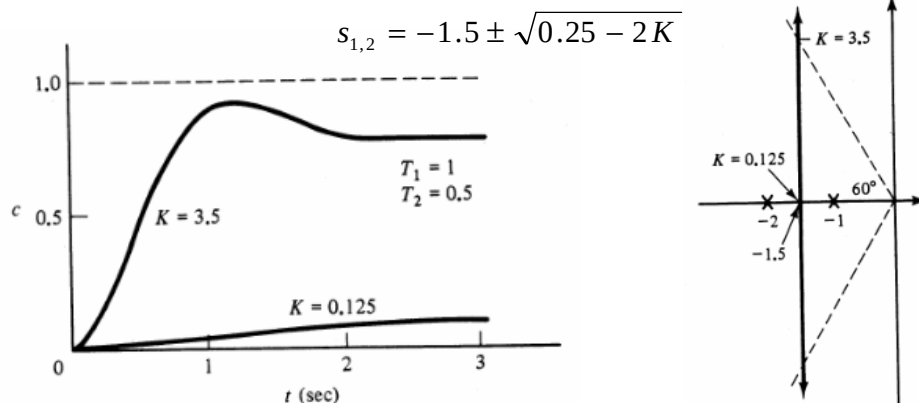
Characteristic equations: $s^2 + 3s + 2 + 2K = 0$

The closed-loop poles are:

$$s_{1,2} = -1.5 \pm \sqrt{0.25 - 2K}$$

Transient Response vs Steady-State Errors

Example-2: P Control with Two Simple Lags



The increase of K that is desirable to increase the speed of response and to reduce steady-state errors, parameter sensitivity, and disturbance response is limited by considerations of relative stability.

Transient Response vs Steady-State Errors

- ◇ Often the maximum value of K permitted by these considerations is not large enough to meet specifications on the steady-state error $1/(1 + K)$.
- ◇ Indeed, in many cases this error is required to be zero. This motivates the use of dynamic compensation.
- ◇ In proportional control, or P control, only gain adjustment is available to improve performance.
- ◇ If performance specifications cannot be met by P control, it must be replaced by a **dynamic controller or dynamic compensator** to provide more flexibility.

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Summary

- Sensitivity is the measure by which controlled signals are influenced by disturbances, which include parameter variations within the plant and external signals such as noise.
- For low sensitivity to plant parameter variations as well as noise, the loop gain should be selected as large as possible.
- High loop gains is also advantageous to reduce the steady-state errors in feedback systems.
- Relative stability considerations usually limit the permissible gain.

Questions?